7. Bats emit high frequency sound waves and receive reflected echoes. They use the echoes to locate their position. This process is called echolocation.

Sound waves emitted by the bat travel at 340 m s^{-1}. Their typical frequency range is 20 kHz to 80 kHz.

(a) Calculate the range of wavelengths for this frequency range. [2]

(b) Bats emit two waveforms, wave B and wave P, which superpose to form wave E.

Wave B (shown in Fig. 7.1) gives information about the surrounding background.

Wave P (not shown in Fig. 7.1) enables the bat to detect insect prey.
Wave E (shown in Fig. 7.1) is the superposition of wave B and wave P.

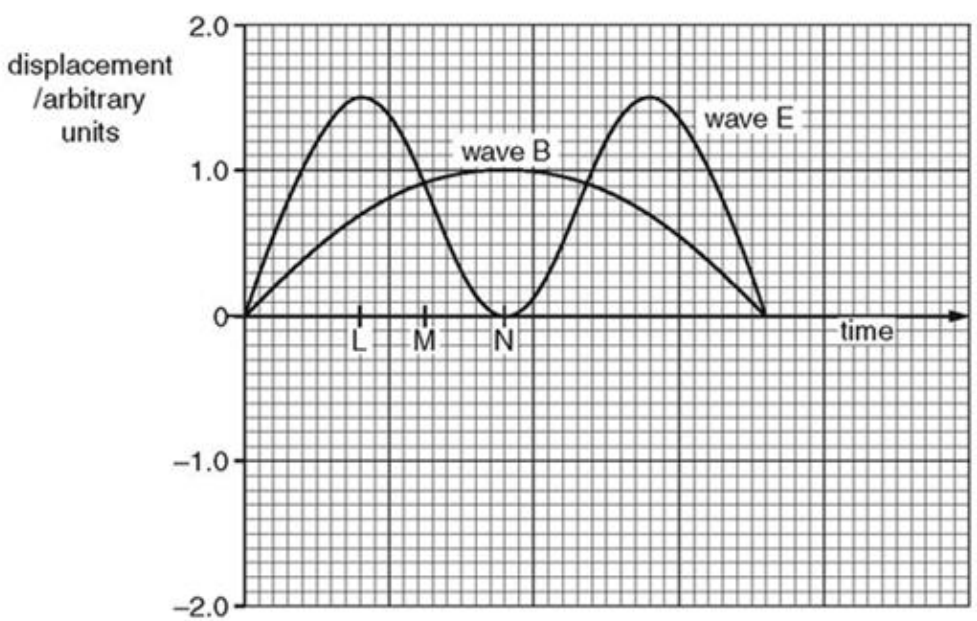

Fig. 7.1

(i) Use the principle of superposition to determine the displacement of wave P at times corresponding to points L, M and N on the time axis.

Write the displacement values in the spaces provided. [3]

displacement of wave P at L = ... units

displacement of wave P at M = ... units

displacement of wave P at N = ... units

(ii) Hence draw the waveform for wave P on Fig. 7.1. [1]

(c) An effect known as the Doppler effect uses changes in frequency to determine speeds.

The change in frequency, Δf, shown by wave P when it is reflected by an insect travelling with speed v is given approximately by the formula

$$\frac{\Delta f}{f} = \frac{2v}{c}$$

where c represents the speed, 340 m s^{-1}, of sound waves emitted by the bat.

(i) Wave P has a frequency of 50.80 kHz. Its apparent frequency after reflection is 51.25 kHz.

Calculate the speed of the insect. [2]

(ii) The bat best discriminates small insect prey when the wavelength of the reflected wave P is similar in size to the insect.

State the wave property that is being demonstrated in this situation. [1]

(d) The bat's high frequency waves are strongly attenuated in air. Fig. 7.2 shows the variation of intensity *I* with range in air *x* for the high frequency waves.

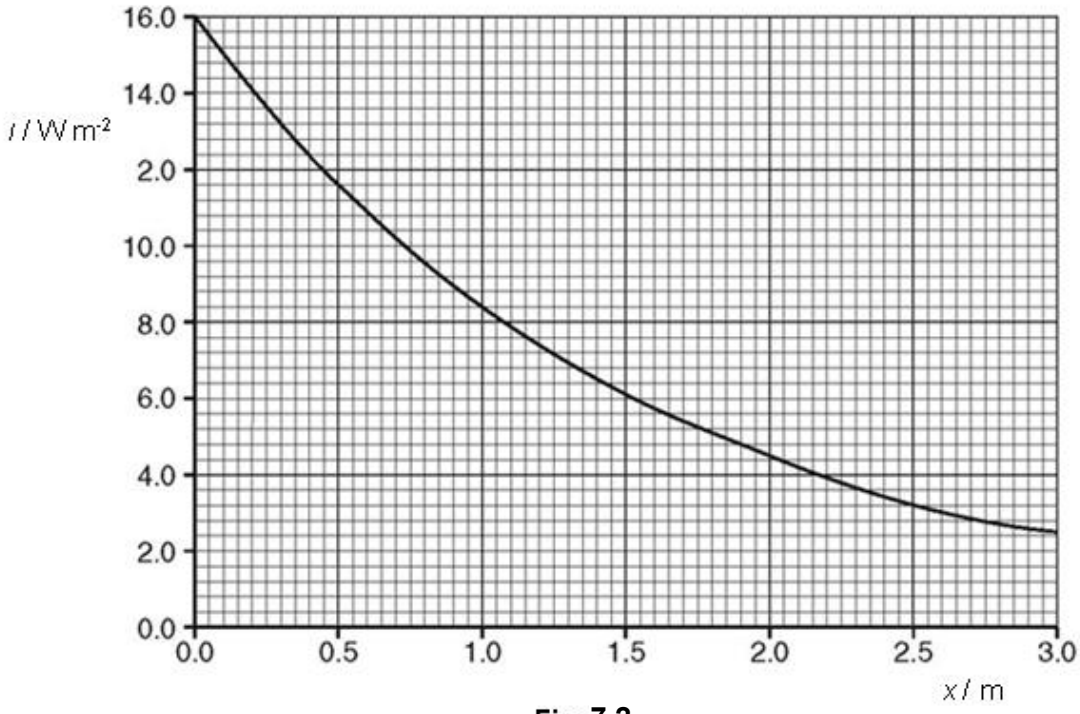

Fig. 7.2

The shape of the curve in Fig. 7.2 suggests that the decrease of the intensity *I* with range in air *x* could be exponential. In order to test this suggestion, a graph of ln *I* against *x* is plotted. This is shown in Fig. 7.3.

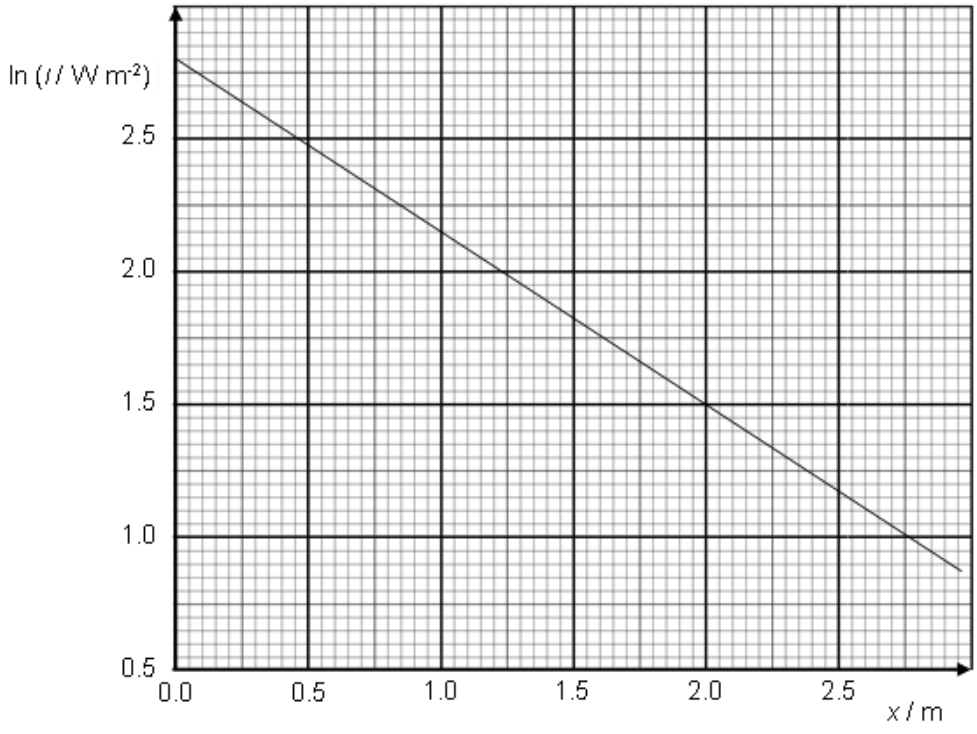

Fig 7.3

(i) Show that Fig. 7.3 indicates a relationship of the form

$$I = I_o e^{-\alpha x}$$

where α is a constant. [3]

(ii) The constant α is known as the attenuation coefficient.

1. State the SI unit of α. [1]

2. Use Fig. 7.3 to show that the value of α is about 0.7 units. [1]

(iii) On the axes of Fig. 7.3, sketch a graph to show a possible variation with range in air *x* of the intensity *I* of this high frequency wave if the value of α is 1.4. [2]

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